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Longitudinal alignments and syntheses of multimodal clinical data for personalized medicine with the PULSE framework

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Multimodal models capable of imputing diverse data used in single-cell biology studies provide potential foundational opportunities in clinical practice. However, patient data uniquely comprise longitudinal mosaic measurements that reflect underlying physiological dynamics and exhibit temporal covariation, demanding a specialized approach. Here we present Patient Unified Longitudinal Signal Engine (PULSE), a longitudinal self-supervised framework that explicitly encodes personalized past states (historical paired modalities) to reconstruct full profiles from subsequent unpaired measurements, thus enhancing current visit multimodal alignment and generation. Applied to the UK Biobank, PULSE accurately generates metabolomic profiles and proteomic profiles from sparse routine blood tests. Compared with the ground truth metabolomic data (251 biomarkers), PULSE-generated profiles outperformed all benchmark methods. Furthermore, the framework accommodates incorporation of retinal images, electronic health records and blood markers with disease prediction: models trained on the generated proteomic profiles achieved areas under the curve of 0.72–0.83 for six common diseases, comparable to that using ground-truth proteomic data. The PULSE framework demonstrates that cross-modal alignment captures the continuous spectrum of disease physiology and extracts robust features that transcend the limitations of traditional binary case–controls.

Precision medicine leverages high-resolution, multimodal measurements to capture human physiological complexity[1–5]. Deep phenotyping, such as high-throughput proteomics and metabolomics, provides rich functional signatures and when collected longitudinally, reveals within-individual changes invisible in cross-sectional data[6–9], while large-scale efforts such as the UK Biobank demonstrate the value of dense profiling[10]. However, the cost and operational complexity of repeated deep multimodal profiling limit its use in routine care[11], limiting access to well-resourced research settings[12,13]. Most healthcare systems therefore still rely on sparse laboratory measurements that capture only fragments of an individual's physiological state.

Single-cell multimodal methods offer a conceptual starting point: distinct modalities (transcriptome, chromatin accessibility and surface proteins) are treated as complementary views of a shared biological state. By leveraging this principle, methods such as MultiVI[14], scVAEIT[15] and MIDAS[16] establish that distinct modalities measured on the same biological entity can be strongly coupled through a shared latent state and can be used to reconstruct one another. However, these methods are typically suited to snapshot observations because each cell usually can be measured once. Clinical data differ fundamentally: the same individual is measured repeatedly, creating longitudinal multimodal data where information lies both within visits (cross-modal dependencies) and across visits (temporal covariation). A test result is more

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informative when interpreted relative to prior measurements and patient-specific baselines[13,17]. Extending single-cell frameworks to clinical data requires explicit modeling of cross-visit temporal context.

Longitudinal clinical modeling has advanced rapidly, yet most applies general sequence architectures (recurrent networks and transformers) to single-modality measurements, such as laboratory values or clinical events recorded over time[18,19]. Multimodal extensions typically rely on input-level concatenation or parallel encoding with late fusion[3,20]. Representation learning often depends on diagnostic labels or case–control outcomes[6,21,22]. However, physiology evolves along a continuous spectrum; discrete labels incompletely capture underlying states and may inherit label bias[23]. Existing approaches therefore fail to model cross-visit and cross-modal relationships or to handle realistic missingness and missing-modality imputation[24].

We propose that individuals should be modeled as dynamic systems where modalities are temporally covarying, complementary observations of an evolving latent physiological state. Under this view, changes in one modality constrain plausible trajectories in others, and prior states enhance missing-modality imputation. We formalized this as the Patient Unified Longitudinal Signal Engine (PULSE), which learns patient-specific temporal context to improve current-visit imputation. We demonstrate PULSE across diverse settings: metabolomic imputation from routine blood tests in the UK Biobank; proteomic generation from laboratory tests plus medical history; 30-day intensive care unit (ICU) readmission prediction jointly modeling laboratory data and clinical text across US and Chinese cohorts; and enhanced cross-body disease prediction through alignment of retinal imaging with blood tests.

Results

The PULSE framework

To model longitudinal multimodal clinical data with incomplete observations, we developed PULSE (Fig. 1a,b). PULSE is based on the premise that different clinical modalities are biologically coupled, temporally covarying readouts of an individual's evolving physiological state, rather than independent targets for direct translation. As patients differ in their baseline physiology and prior trajectory, the same measurement at a given visit may correspond to different underlying states. Thus, the physiologically informative signal lies not only in the absolute value observed at the current visit but in its change from the patient's prior state, which in turn constrains plausible changes in other modalities.

PULSE operationalizes these principles by explicitly carrying forward an individualized latent summary of the previous visit, termed the Past-State, and using it as context for modeling the current visit. At visit N, PULSE encodes each observed modality into a modality-specific latent representation, while a dedicated Past-State branch summarizes patient-specific historical physiology from visit $N-1$. These representations are mapped into a shared latent space through soft distributional alignment, which promotes physiological consistency across modalities and across time without forcing them into an identical code. This encourages PULSE to capture cross-visit shared information that reflects an individual-specific physiological state, while still allowing temporal variation around that state. The aligned signals are integrated into a unified Visit-State, which is decoded to reconstruct all modalities at the current visit, including those not observed, thereby enabling cross-modal imputation and placing incompletely observed visits in the same physiological space as fully observed visits. This reconstruction supervisory anchor ensures that the current-modality latent states remain faithful to the present observations, while the historical state serves only as supportive context for the current state rather than a template that the current data must follow. We also show theoretically that conditioning on the Past-State yields a more informative and tighter posterior over the current latent state, thereby reducing the posterior reconstructive variance for missing modalities (Supplementary Note 1).

During training, we applied an active modality masking strategy to encourage true cross-visit and cross-modal inference. Randomly selected observed modalities were masked from the encoder inputs but retained as reconstruction targets, forcing the model to reconstruct them from the remaining modalities and the patient-specific historical context encoded in the Past-State. Reconstruction losses were applied wherever ground truth was available, while adversarial objectives reduced artifacts arising from missingness patterns. The Visit-State was then propagated through a temporal module, implemented in either recurrent neural network (RNN)-like or transformer-like form, to generate the Past-State for visit $N+1$ and model patient trajectories as individualized physiological state transitions rather than disconnected snapshots.

Benchmarking PULSE against state-of-the-art multimodality methods

We benchmarked PULSE using ~10,000 UK Biobank participants who had paired measurements from one baseline visit and one follow-up visit across two modalities: 61 routine laboratory tests and 251 metabolomics biomarkers (Supplementary Table 1). Individuals were randomly assigned at the patient level to a training set (70%, $n = 7{,}000$) and a held-out test set (30%, $n = 3{,}000$), such that all visits from each test individual were completely excluded from model training. To ensure a fair comparison, all methods were evaluated under the same patient-level train/test split and on the same reconstruction target. The evaluation task was to reconstruct Follow-up Visit metabolomics from Follow-up Visit laboratory tests. For PULSE, the same individual's baseline visit full multimodal profile was additionally used to provide historical context, consistent with the intended longitudinal use of the framework.

As baselines, we adapted three representative single-cell multimodal methods spanning distinct integration paradigms (see the 'Adaptation of single-cell multimodal frameworks' section in the Methods for a description of the adaptation procedures): MIDAS, based on variational inference with a PoE architecture[16,25]; scVAEIT, which uses a probabilistic variational autoencoder to learn a joint latent space[14,15]; and StabMap, which performs cross-modality alignment through mutual nearest neighbors[26,27]. Notably, we treated them as adapted static multimodal frameworks: each visit from training-set individuals was treated as an independent paired sample, and the models were trained without access to historical-state variables. In the test, all static frameworks received only follow-up visit laboratory tests as input and were evaluated on reconstruction of the corresponding Follow-up Visit metabolomics, thereby matching the reconstruction target used for PULSE. All formal benchmark comparisons were performed across the complete, unfiltered panel of 251 metabolites. For each metabolite and each method, we computed Pearson's r, mean absolute error (MAE) and mean squared error (MSE) between predicted and measured values in the held-out test set. Under this evaluation, PULSE consistently outperformed all benchmark methods across all three metrics (Fig. 2a and Supplementary Table 2; $P < 0.001$).

We next examined which metabolites were reconstructed most faithfully by PULSE. To provide a feature-level view, we highlighted the ten metabolites with the highest Pearson correlations between predicted and measured values in the full PULSE model (Fig. 2b). Notably, many of the top-ranked features were lipid ratio markers, which are physiologically important and often thought to reflect relatively stable aspects of metabolic regulation. This observation prompted us to ask whether differences in metabolite predictability might be related to their longitudinal stability. Motivated by recent studies suggesting that many blood markers fluctuate around stable, patient-specific set points rather than a single population average[28,29], we investigated whether variation in imputation performance could be explained by intra-individual stability. Specifically, for each metabolite, we quantified intra-individual stability by computing the Spearman correlation coefficient between two longitudinal measurements across

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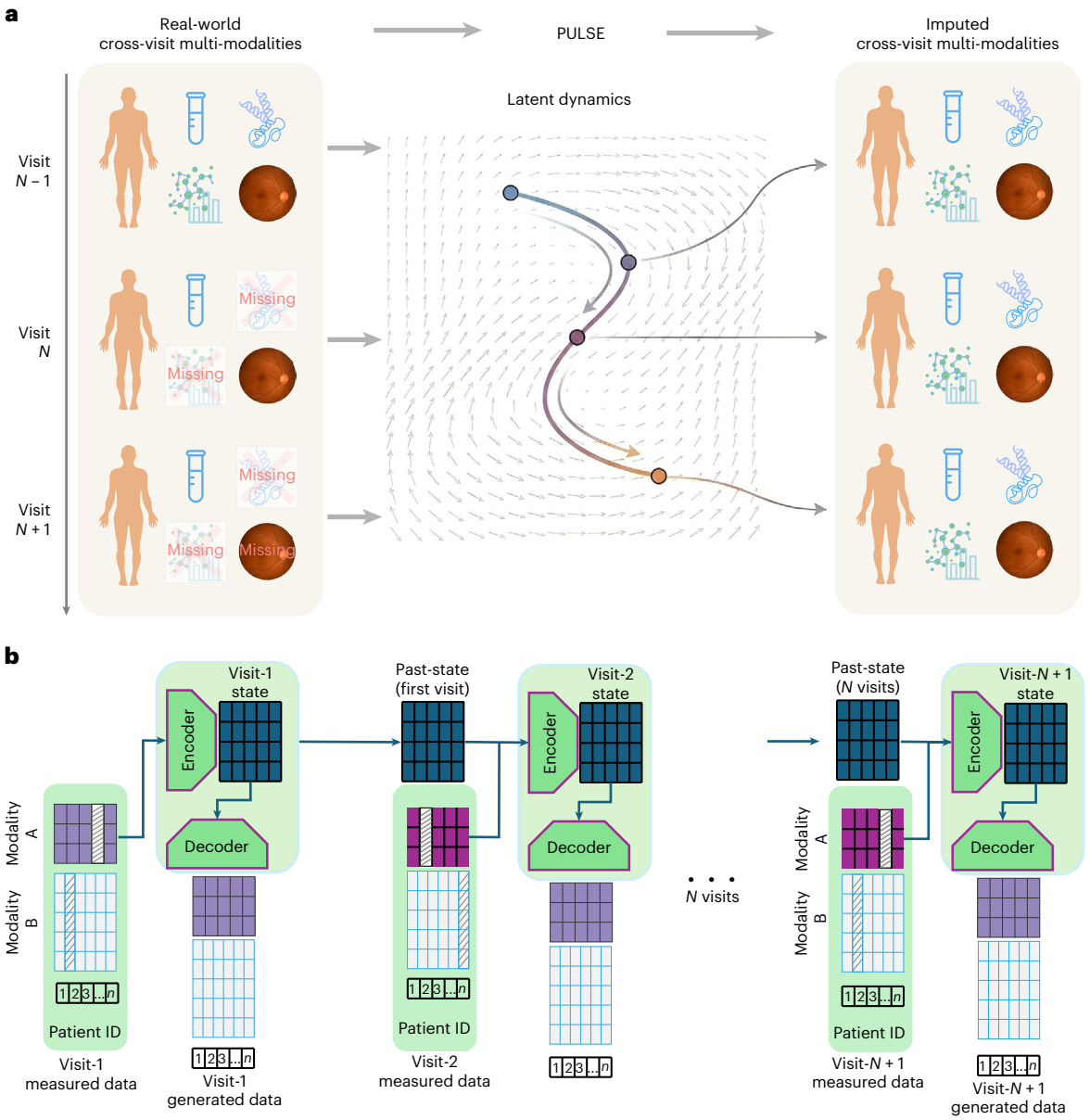


Fig. 1 | The PULSE framework. a, Conceptual schematic of real-world longitudinal multimodal measurements with missing modalities at different visits (left, crossed-out icons). Modalities are treated as temporally covarying readouts of an evolving physiological state; the framework uses this structure to infer or impute a complete multimodal description at each visit (right). **b**, Architecture of the PULSE framework. Encoders integrate the latent 'Past-State' from previous visits with current sparse 'Measured Data' across modalities to derive a unified 'Visit State'. The Past-State vector produced at the previous visit is not reused verbatim: it is passed through a dedicated Past-State encoder at the current visit to obtain a historical-context latent branch that is aligned and fused with the current-visit branches into a unified Visit-State. Decoders use this joint latent representation to generate complete modalities. A recurrent mechanism propagates the latent state from the previous visit (for instance, Visit-*N* State) to serve as the Past-State input for the subsequent visit (for instance, Visit-*N* + 1).

all individuals with paired data available. This metric assesses the consistency of individual differences over time: a high correlation indicates stable individual set points under physiological homeostasis. As a result, metabolites with higher intra-individual stability were predicted with significantly greater accuracy (lower MAE and MSE and higher Pearson's *r*) (Supplementary Fig. 1). This relationship suggests that PULSE's imputation performance is biologically grounded. Metabolites under strict homeostatic regulation are inherently more predictable because they reflect consistent underlying physiological states. Conversely, metabolites with lower intra-individual stability, which may lack specific set points and fluctuate more readily with short-term perturbations, present greater challenges for longitudinal prediction. This result clarifies why certain biomarkers achieve higher reconstruction fidelity than others and supports the physiological plausibility of PULSE's predictions.

As a clinically relevant example, we further examined the free cholesterol-to-total lipid ratio in large LDL particles, a biomarker associated with multiple diseases in the UK Biobank. For this feature, we compared predicted longitudinal change (generated follow-up value minus baseline value) with measured longitudinal change (ground-truth follow-up value minus baseline value), as the change over time is often more clinically informative than the absolute level alone. Full PULSE showed the strongest agreement between predicted and measured change ($r = 0.850$), outperforming all static frameworks (Fig. 2c).

We next performed a series of ablation analyses to examine the contribution of the major design choices in PULSE. Removing the

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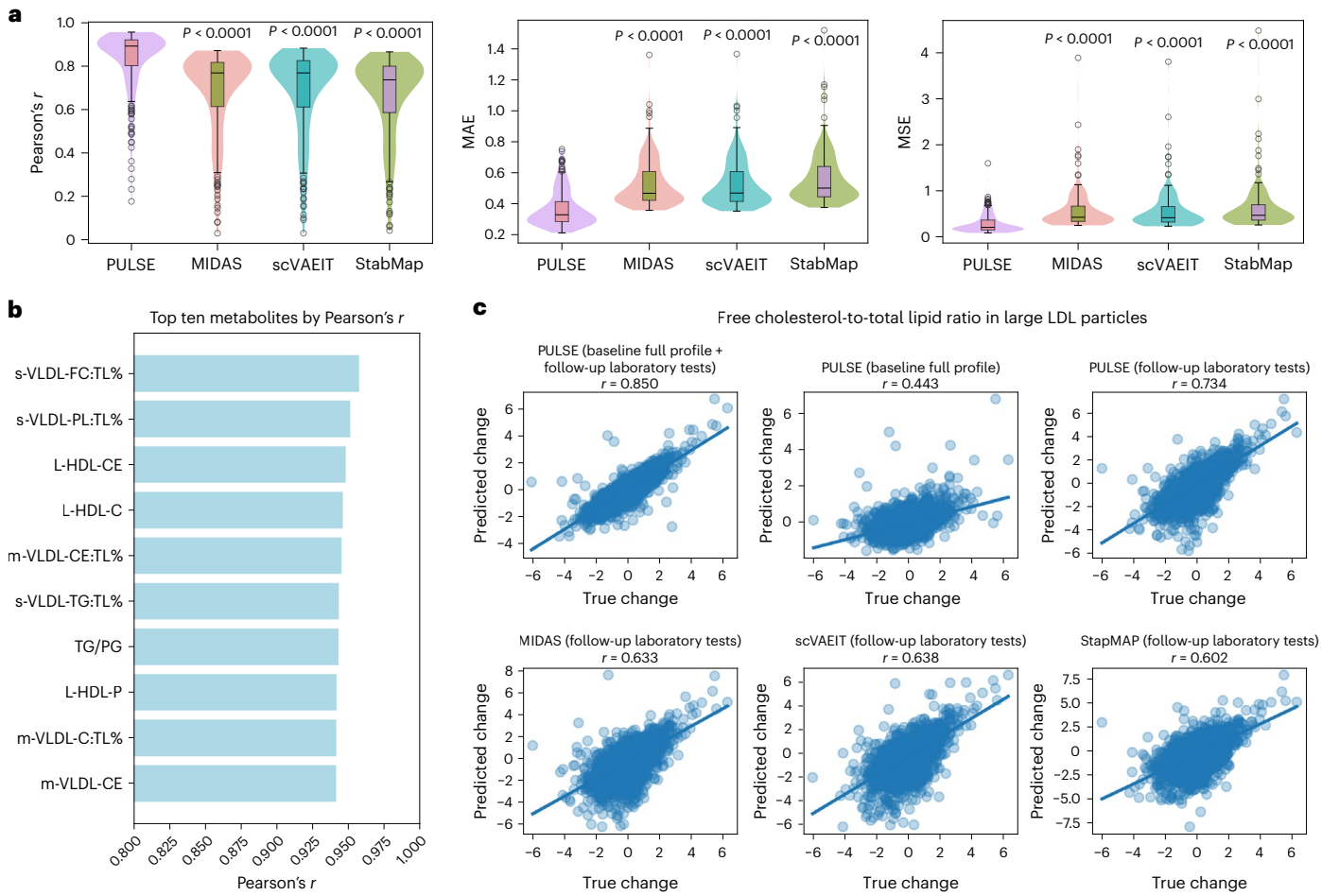


Fig. 2 | Benchmarking PULSE performance on longitudinal metabolomics imputation. a, Quantitative comparison of imputation accuracy for 251 metabolomics biomarkers across the test set ($n = 3{,}000$) using Pearson's r (left), MAE (center) and MSE (right). Box plots display performance distributions for PULSE and three static baseline methods (MIDAS, scVAEIT and StapMap). Center lines indicate medians; box limits indicate the 25th and 75th percentiles; whiskers extend to 1.5 times the interquartile range; points indicate outliers. For each pair of models (PULSE versus comparator), we compared metabolite prediction performance (Pearson's r, MAE and MSE, $n = 251$) using a two-sided Wilcoxon signed-rank test. Exact P values are shown for $P \geq 0.0001$; values below this threshold are denoted as $P < 0.0001$. No adjustment for multiple comparisons was applied. **b**, Exploratory illustration: bar chart showing the top ten metabolites with the highest reconstruction fidelity, ranked by Pearson's r from the full PULSE model. The metabolites are defined below, with their corresponding UK Biobank data field IDs given in parentheses: s-VLDL-FC:TL%, free cholesterol to total lipids percentage in small VLDL particles (field ID 23602); s-VLDL-PL:TL%, phospholipids to total lipids percentage in small VLDL particles (field ID 23599); L-HDL-CE, cholesteryl esters in large HDL particles (field ID 23562); L-HDL-C, cholesterol in large HDL particles (field ID 23561); m-VLDL-CE:TL%, cholesteryl esters to total lipids percentage in medium VLDL particles (field ID 23596); s-VLDL-TG:TL%, triglycerides to total lipids percentage in small VLDL particles (field ID 23603); TG/PG, triglycerides to phosphoglycerides ratio (field ID 23435); L-HDL-P, phospholipids in large HDL particles (field ID 23560); m-VLDL-C:TL%, cholesterol to total lipids percentage in medium VLDL particles (field ID 23595); m-VLDL-CE, cholesteryl esters in medium VLDL particles (field ID 23506). **c**, Scatter plots illustrating the concordance between ground-truth and predicted longitudinal changes (defined as follow-up value minus baseline value) for a representative biomarker, free cholesterol-to-total lipid ratio in large low-density lipoprotein (LDL) particles. In each scatter plot, the displayed r denotes the Pearson's r between the predicted longitudinal changes and the ground-truth changes.

historical-state input substantially reduced reconstruction performance, confirming that patient-specific prior context provides nonredundant information for follow-up metabolomics imputation (Supplementary Fig. 2). To directly address the concern that alignment might inappropriately pull current-visit representations toward the past, we evaluated a directional stop-gradient variant in which the current-visit modality representations were treated as fixed references and only the historical branch received alignment gradients. This design explicitly prevents the current state from being optimized toward the historical state. We also tested a stricter hard point-wise alignment variant. Neither alternative improved performance over the default alignment, and the hard point-wise strategy led to a decline in reconstruction accuracy (Supplementary Fig. 3). This empirical result is consistent with our theoretical analysis (Supplementary Note 2), which demonstrates that for the task of imputation under partial observability, preserving the full latent posterior through distributional alignment is a more principled route toward Bayes-optimal reconstruction than deterministic point-wise mapping. Finally, loss-level ablations showed that removing the alignment loss caused a clear performance drop, whereas removing the adversarial loss produced a more modest but still negative effect (Supplementary Fig. 4). Together, these results support the historical-state conditioning and chosen alignment mechanism as core contributors to PULSE performance, with the adversarial component providing an additional complementary benefit.

PULSE-inferred clear aging trajectory in joint latent space

To investigate the biological structure captured by the PULSE joint latent space, we constructed an artificially unpaired dataset consisting

of 50% fully paired samples, 25% laboratory-test-only samples and 25% metabolomics-only samples. PULSE was trained solely on the paired subset and inferred latent representations for all samples. Using Uniform Manifold Approximation and Projection (UMAP) visualization of the joint latent space, we observed that despite incomplete inputs, all three sample types were embedded into a coherent shared manifold (Fig. 3a,d), confirming PULSE's ability to align partially observed modalities into a unified physiological representation.

We next investigated the biological variation encoded within this UMAP space. When the manifold was colored by chronological age, a clear global gradient emerged along the UMAP1 axis, with younger individuals concentrated on the right and older individuals shifted toward the left (Fig. 3b,e). However, we observed substantial heterogeneity that was not captured by chronological age alone: individuals within the same narrow age strata exhibited marked dispersion along the UMAP2 axis, forming distinct 'upper', 'middle' and 'lower' branches. This suggested that rather than following a single, uniform aging path, individuals might be traversing multiple aging-related directions.

To test whether these divergent branches observed in UMAP space reflect true long-range trajectory structures rather than static or local projection artifacts, we performed trajectory inference on the high-dimensional latent representations using a CellRank-based trajectory inference workflow[30,31] (see the 'Biological aging trajectories and grouping strategy' section in the Methods). A vector field of transition probabilities was computed in the latent representations; streamlines tracing long-range directional flows were derived from this field and projected onto the UMAP space for visualization. The resulting streamline field, projected onto UMAP space for visualization, demonstrated three consistent long-range directional flows: samples in the upper UMAP2 region progressed toward the upper-left, middle-region samples toward the central-left and lower-region samples toward the lower-left. This confirms that the vertical (UMAP2) axis maintains directional consistency across the full manifold—the 'upper/middle/lower' branches reflect genuine trajectory topology in the latent space, not UMAP projection artifacts.

To explore the clinical importance of this heterogeneity, we stratified participants into three trajectory groups on the basis of their relative UMAP2 positions within narrow age strata (for instance, grouping individuals aged 50–51 years by their UMAP2 coordinates and repeating across the age range). Pooling these assignments yielded three definitive trajectory groups (Fig. 3c,f). We then conducted a cumulative disease risk analysis for major age-related conditions (Fig. 3g,h). The analysis revealed that the 'red group' consistently exhibited the highest cumulative incidence of chronic obstructive pulmonary disease (COPD), diabetes, heart failure and stroke compared with their age-matched peers in other groups. Consequently, we interpret this specific branch as representing an accelerated-aging trajectory.

Leveraging clinical context to generate actionable molecular profiles for disease prediction

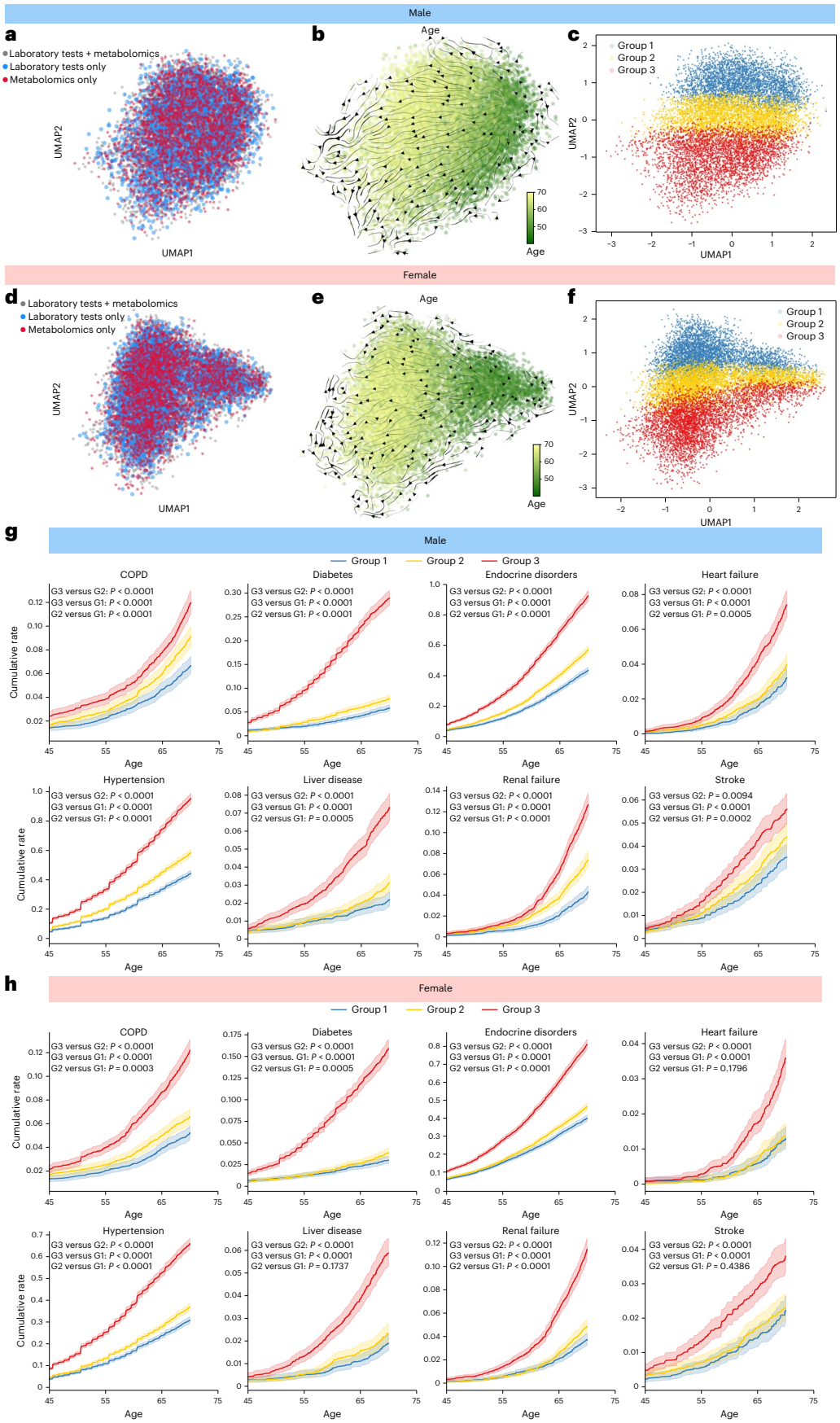
Although our core framework leverages a patient's prior multiomics state as context, such data are rarely available in routine care. By contrast, medical history, encompassing diagnoses, medications and comorbidities, is routinely recorded and provides a rich longitudinal summary of a patient's physiological trajectory. Therefore, we hypothesized that this easily accessible clinical context could serve as a powerful surrogate to guide the generation of expensive molecular profiles, such as plasma proteomics, from inexpensive routine laboratory tests (Fig. 4a). To test this, we applied PULSE to the UK Biobank-PPP cohort (n = 50,000; Supplementary Table 1), which includes paired proteomic (2,923 features), laboratory test (61 features) and medical history data. We split the cohort at the patient level into a 70% development set for training and a 30% held-out test set for evaluation. The task was to generate proteomics profiles of a visit using laboratory tests from that visit, combined with the encoded medical history context.

We benchmarked the PULSE model against three adapted single-cell methods (MIDAS, scVAEIT and StabMap) and PULSE ablations. The PULSE model was trained on the development set using paired routine laboratory tests, encoded medical history and measured proteomics and evaluated on the held-out test set by generating proteomics from laboratory tests and medical history. The single-cell baseline methods were trained on paired laboratory tests and proteomics in the training set but evaluated using laboratory tests only (without medical history). Performance was quantified using Pearson's correlation (r) between generated and measured protein abundance for each protein in the held-out test set. For direct comparison, we plotted each method's per-protein r against that of the full PULSE model (Fig. 4b); nearly all points fell below the diagonal ($y = x$), indicating that PULSE outperformed the benchmarks for the majority of proteins. We additionally computed Spearman rank correlations between generated and measured levels as more robust analysis; the same pattern held, with nearly all points below the diagonal (Supplementary Fig. 5). We further examined performance on a clinically important subset of 15 pleiotropic proteins (Fig. 4c). For these proteins, the full PULSE model also achieved a high Pearson's r and outperformed all other baselines.

We further benchmarked PULSE against established approaches for learning patient trajectories from large-scale electronic health record data on the proteomics generation task. We considered three representative paradigms: (1) Delphi[18], a transformer-based model for longitudinal diagnosis-event trajectories that does not incorporate routine laboratory tests as inputs, adapted here to predict proteomics directly from its trajectory representation; (2) adapted general large language model (LLM) (Llama 3.1)[20], by converting medical history and routine laboratory tests into natural-language prompts for a general LLM, from which we derived text embeddings for downstream proteomics prediction; and (3) a standard Transformer encoder that tokenizes disease codes and laboratory values for direct sequence modeling, trained end-to-end to predict proteomics. PULSE significantly outperformed all three baselines in proteomics generation accuracy, yielding a clear overall ranking (PULSE > Transformer > Llama 3.1 > Delphi; Supplementary Fig. 6). Delphi's weaker performance is consistent with its limited use of concurrent laboratory measurements, which are informative for proteomic inference. Llama 3.1 leverages rich contextual narrative but can dilute fine-grained quantitative

Fig. 3 | Visualization of the PULSE joint latent space and its association with aging and disease risk. a,**d**, Two-dimensional UMAP of PULSE latent follow-up visit states for men (**a**) and women (**d**). The point color indicates input availability at follow-up: paired laboratory tests and metabolomics (gray), laboratory tests only (blue) and metabolomics only (red). **b**,**e**, The same UMAP coordinates for men (**b**) and women (**e**) as in **a** and **d**, colored by chronological age at the visit (color bar, years). Streamlines show the transition structure from the CellRank trajectory inference computed in high-dimensional latent space and projected onto UMAP (Methods). **c**,**f**, Partitioning of the joint latent space into three trajectory groups (group 1: blue, group 2: yellow, group 3: red) on the basis of UMAP2 coordinates within narrow age strata for men (**c**) and women (**f**). **g**,**h**, Cumulative incidence plots for eight major age-related diseases in men (**g**) and women (**h**). The x axis represents age (years), and the y axis represents the cumulative rate of disease incidence. The center line represents the cumulative hazard, computed per group as the cumulative sum of discrete hazards derived from the Kaplan–Meier survival estimate. Shaded areas matching each curve color denote the 95% CIs of the cumulative hazard estimates. These CIs were derived from the CIs of the original Kaplan–Meier survival function estimates. Pairwise group differences were assessed with two-sided log-rank tests comparing survival distributions by age at onset/censoring; uncorrected P values are indicated for G3 versus G2, G3 versus G1 and G2 versus G1. No adjustment for multiple comparisons was applied across the three pairwise tests per panel. For P values ≥0.0001, we present exact P values. P values <0.0001 we denote as $P < 0.0001$.

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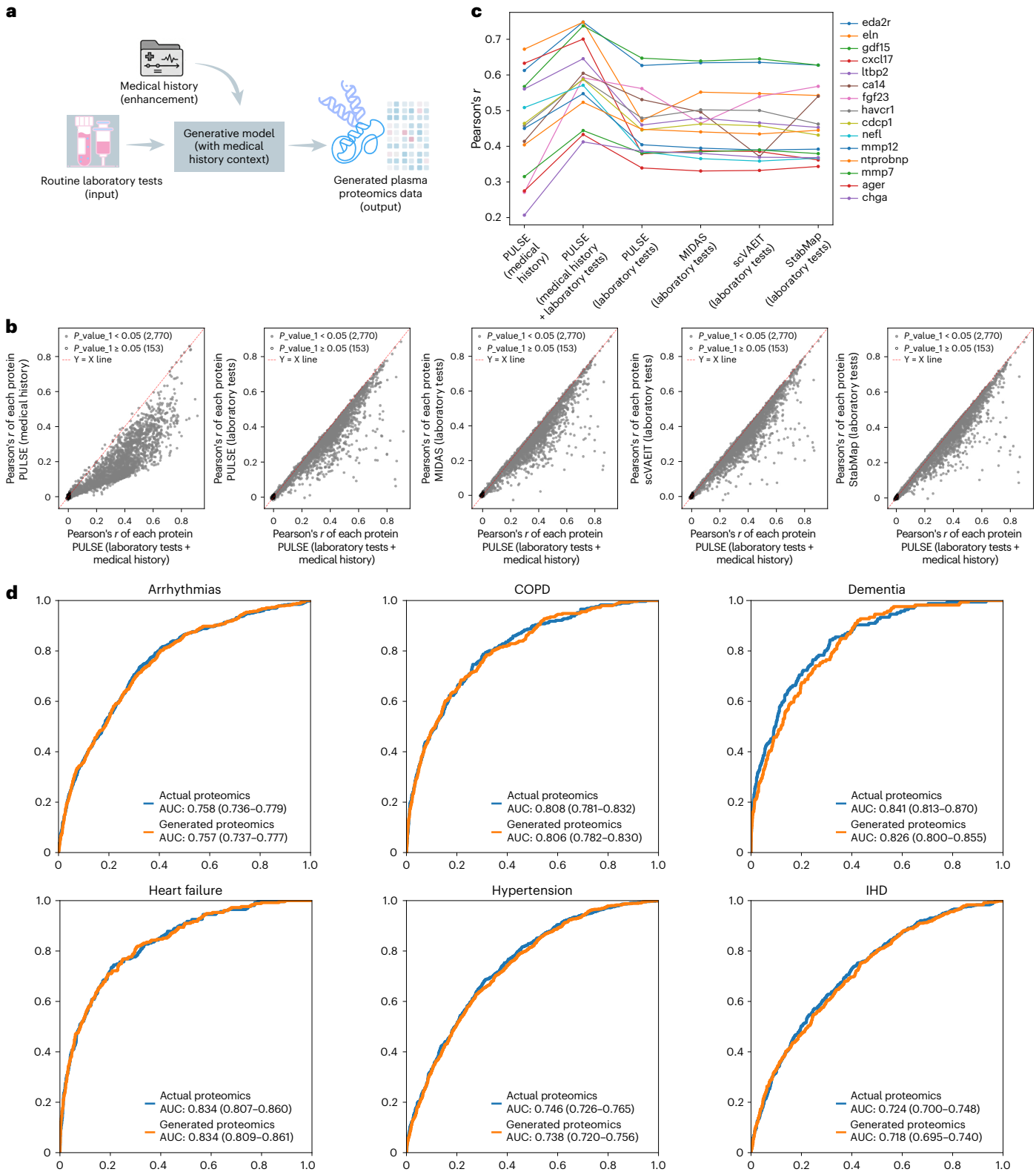


Fig. 4 | Generation of plasma proteomics from routine clinical data and medical history. a, Schematic illustration of PULSE leveraging routine laboratory tests and medical history context to generate high-dimensional plasma proteomics. **b**, Performance comparison of protein reconstruction across the held-out test set ($n = 15{,}000$). Scatter plots display the Pearson's r for each protein between generated and ground-truth values. The x axis represents the r achieved by the full PULSE model, while the y axis represents the r for ablated PULSE versions or baseline methods (MIDAS, scVAEIT and StapMap). Points below the $y = x$ diagonal indicate better performance of the full PULSE model. **c**, Reconstruction accuracy (Pearson's r) for 15 disease-related pleiotropic proteins across different models and input configurations. **d**, ROC curves for predicting six major age-related diseases using actual proteomics (blue) versus PULSE-generated proteomics (orange). AUC values with 95% CIs are provided. IHD, ischemic heart disease; ROC, receiver operating characteristic.

structure during text conversion. The standard Transformer uses the available structured inputs end to end but lacks an explicit mechanism for cross-modal alignment that jointly encodes a coherent physiological state.

To assess the clinical utility of generated versus measured proteomic profiles for disease prediction, we evaluated both in six major age-related diseases. On the development cohort (Methods), we trained Light Gradient Boosting Machine (LightGBM) models—a standard choice for proteomics-based prediction—using measured proteomics with age and sex as the reference inputs and evaluated predictions on the held-out test cohort. We then repeated training and evaluation using PULSE-generated proteomics (with age and sex) and directly compared performance with the measured-proteomics models. The area under the receiver operating characteristic curve (AUROC) was comparable between generated and measured inputs across all six diseases (Fig. 4d). As the area under the precision–recall curve (AUPRC) is often more informative under class imbalance, we additionally reported AUPRC and included an age-and-sex-only LightGBM baseline. Models trained on generated proteomics achieved a AUPRC comparable to measured proteomics across diseases, with both substantially outperforming the demographic baseline (Supplementary Fig. 7). Individual risk scores were also tightly aligned between the two settings (Supplementary Fig. 8a), indicating that generation preserves biological signal sufficient for risk stratification. We also performed a test on the remaining UK Biobank participants (n = 400,000) without measured proteomics. A model using laboratory tests, medical history and PULSE-generated proteomics significantly outperformed a model using only the original laboratory tests and medical history (Supplementary Fig. 8b). This result suggests that the generative process is a potent representational optimizer. It effectively distills information from the sparse, heterogeneous clinical inputs into a structured and biologically richer proteomic feature space.

To further test whether comparable accuracy between generated and measured proteomics reflected genuine biology rather than statistical artifacts, we performed Shapley Additive Explanations (SHAP)-based interpretability on LightGBM models trained separately on each input type and compared the top important protein. Across diseases, the most influential proteins were highly concordant between the two input settings (Supplementary Fig. 9), supporting that generated proteomics preserves clinically meaningful structure and matches measured proteomics not only at the level of global performance metrics but also at the level of feature-level explanations.

Multicenter validation of PULSE for heterogeneous longitudinal multimodal modeling

To further demonstrate that PULSE can flexibly adapt to varied clinical settings with heterogeneous longitudinal multimodal data, and to evaluate its advantage over representative longitudinal clinical modeling baselines, we applied the framework to two distinct ICU cohorts: the US-based MIMIC-III cohort, comprising 44,265 adult patients (≥18 years old) with 3,025 cases of 30-day ICU readmission, and a China-based large-scale ICU dataset (the Wenzhou ICU cohort), which included 13,816 adult patients (≥18 years old) with 793 cases of 30-day ICU readmission. We focused on predicting 30-day readmission among ICU patients, a complex task requiring the integration of structured laboratories and unstructured clinical notes (see the 'Multicenter 30-day readmission among ICU patients' section in the Methods). We compared PULSE with three representative longitudinal clinical data modeling strategies: (1) a Transformer using the laboratory-test modality only for patient representation, modeling repeated structured variables over time; (2) a general LLM using the text modality only, where all note text within the observation window was encoded by Llama 3 to obtain a patient representation; and (3) a late-fusion model (Directly Combined) that learned separate encoders for structured trajectories and text embeddings and combined the two representations at prediction.

We evaluated all methods on 30-day postdischarge readmission, using only data from the first 48 h after the first ICU intime. For PULSE, this window was divided into 4-h steps. Structured measurements were summarized at each step, and contemporaneous clinical notes were aggregated into step-level text inputs, encoded by Llama 3 and aligned with structured features at each step while propagating patient-level temporal context across steps. All baselines were trained and evaluated under the same input scope, time window, prediction task and train/validation/test splits within each cohort.

In the MIMIC-III cohort, PULSE achieved an AUROC of 0.713 (95% confidence interval (CI) 0.691–0.732) and an AUPRC of 0.193 (0.167–0.224) (Supplementary Fig. 10), outperforming Directly Combined (0.663; 0.130), LLM (clinical notes) (0.647; 0.127) and Transformer (laboratory test) (0.639; 0.110). Similarly, in the Wenzhou ICU cohort, PULSE achieved an AUROC of 0.735 (0.683–0.789) and an AUPRC of 0.388 (0.298–0.480), outperforming Directly Combined (0.706; 0.317), LLM (clinical notes) (0.691; 0.273) and Transformer (laboratory test) (0.647; 0.116). These results support the flexible use of PULSE in acute care and its consistent gains over representative longitudinal baselines on structured- and text-based multimodal ICU trajectories across two healthcare systems.

Multimodal alignment enhances single-modality representation

We extended PULSE to more heterogeneous modalities, retinal fundus images and laboratory tests, to test whether longitudinal multimodal alignment could enhance the predictive power inherent to each individual modality. We hypothesized that, through cross-modal pretraining, PULSE would learn to encode systemic health information within fundus image features and, conversely, capture ocular-relevant signals from laboratory tests. To validate this, we designed a rigorous test of cross-body prediction task: forecasting pediatric myopia progression from laboratory tests and predicting systemic disease from retinal fundus images.

For the myopia progression prediction task, we used a longitudinal cohort of 24,605 children (50,043 visits; Supplementary Table 1) where each participant contributed at least one visit with laboratory tests and one visit with fundus images, although individual visits may contain laboratory tests only (19.4% of visits), fundus images only (15.0%) or both modalities (65.6%). The dataset was randomly partitioned by individual in a 7:3 ratio for development and held-out test sets. We first pretrained PULSE on the development set to align fundus-derived features with laboratory tests across visits. The PULSE latent representation was then fine-tuned, using only laboratory test inputs, to predict future axial length (AL) and spherical equivalent (SE) trajectories over 1–5-year horizons. Against an end-to-end baseline (see the 'Baseline methods of multimodal alignment' section in the Methods) trained solely on laboratory tests, the PULSE pretrain–fine-tune strategy consistently achieved stronger predictive capacity for both AL and SE at every horizon (Fig. 5a–c and Supplementary Fig. 11), demonstrating that cross-modal pretraining distills ocular-relevant structure into laboratory representations that improves longitudinal vision forecasting.

The systemic diseases prediction task was conducted on another longitudinal cohort (Systemic Diseases Prediction Cohort) of 127,731 individuals (657,103 visits) (Extended Data Table 1 and Supplementary Table 1), which was randomly partitioned at individual level as 7:3 into development and held-out test sets. PULSE was first pretrained on the longitudinal 'mosaic' dataset where each visit may contain fundus imaging, laboratory tests or both. This alignment stage produces a align-pretrained fundus encoder whose latent representation already embeds systemic information. We then fine-tuned this encoder for 5-year incident ischemic stroke, myocardial infarction, heart failure and dementia prediction under two regimes (Fig. 5d): (1)

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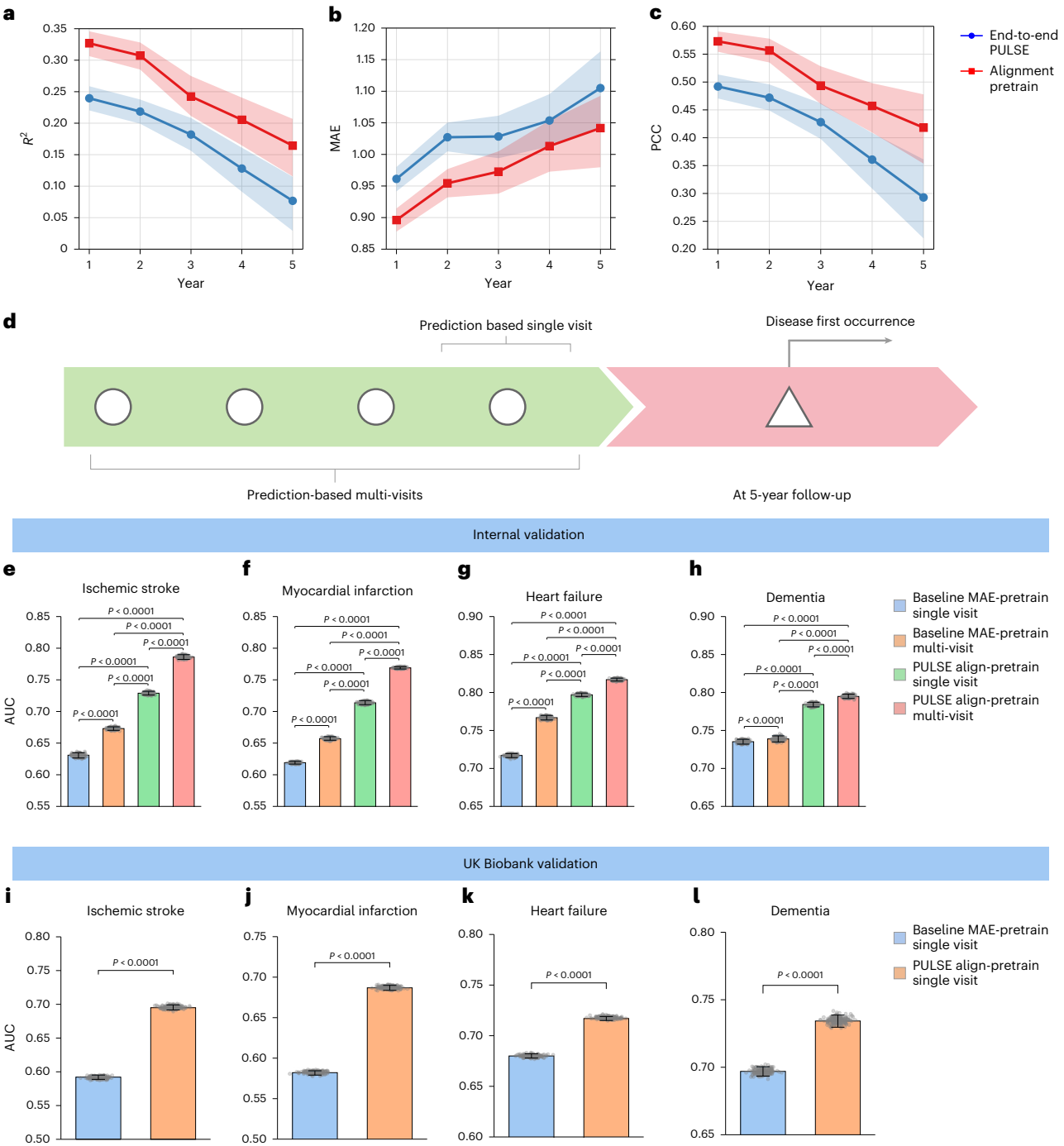


Fig. 5 | Enhancement of single-modality representations through cross-modal alignment. a–c, Performance of myopia progression prediction over 1–5-year horizons using laboratory tests. Comparison between the PULSE align-pretrain strategy (red) and a traditional end-to-end baseline (blue) for predicting AL (left eye) using the coefficient of determination (R^2) (**a**), MAE (**b**) and Pearson correlation coefficient (PCC) (**c**). The comparator baseline end-to-end model: recurrent network applied directly to longitudinal laboratory tests to predict the same targets, without fundus–laboratory alignment pretraining. Solid lines denote mean performance; shaded areas indicate 95% CIs. **d**, Schematic of the prediction regimes for systemic disease risk: single visit (current data only) and multivisit (aggregating longitudinal history). **e–h**, Internal validation for predicting ischemic stroke (**e**), myocardial infarction (**f**), heart failure (**g**) and dementia (**h**) from retinal fundus images. Bars compare the MAE-pretrained RETFound baseline (blue and orange) against the PULSE align-pretrained model (green and pink) under both regimes. Baselines use the MAE-pretrained RETFound encoder: the fundus image at each visit is encoded to a feature vector; in single-visit mode, that vector feeds the disease prediction head; in multivisit mode, visit-wise vectors are aggregated with a recurrent network before disease prediction head. **i–l**, External validation on the UK Biobank cohort ($n \approx 400{,}000$) for the same four diseases using retinal fundus images in a single-visit setting, comparing the RETFound baseline (blue) and PULSE align-pretrained model (orange). Data are presented as mean values, and error bars denote the 95% CIs (2.5th–97.5th percentiles), estimated from 200 bootstrap resamples. The gray dots represent the individual AUROC values obtained in each bootstrap iteration. For each disease, all pairwise comparisons between models were performed. Comparisons used a two-sided Wilcoxon signed-rank test. No adjustment for multiple comparisons was applied. For $P \geq 0.0001$, the exact two-sided P value is denoted. For $P < 0.0001$, $P < 0.0001$ is denoted. In the Systemic Diseases Prediction Cohort, four models were compared (six pairwise tests per disease). In the UK Biobank cohort, two single-visit models were compared (one pairwise test per disease).

a single-visit setting, where only the current fundus image is processed to yield the latent representation that feeds the prediction head, and (2) a multivisit setting, where fundus images from the current and all previous visits are sequentially passed, so that the final visit latent representation aggregates the longitudinal trajectory before prediction. The baseline RETFound pipeline mirrors this structure but starts from masked autoencoding-pretrained (MAE-pretrained) RETFound encoder (see the 'Baseline methods of multimodal alignment' section in the Methods). We fine-tuned MAE-pretrained on the identical disease tasks with the same prediction head, again under single-visit and multivisit scenarios: either the current fundus alone or the ordered series of current plus historical fundus images processed through the recurrent stack. Both models were evaluated on the held-out test cohort in each regime for a fair comparison, the align-pretrained PULSE model demonstrated better predictive capability across all four diseases, achieving higher area under the curve (AUC) values than MAE-pretrained RETFound in both single-visit and multivisit prediction scenarios (Fig. 5e–l). Crucially, the enhanced performance in multivisit predictions demonstrates that longitudinal data integration further strengthens cross-modal disease prediction (Fig. 5e–h). To confirm the generalizability, we employed the UK Biobank dataset as external testing cohort to validate the align-pretrained PULSE model and MAE-pretrained RETFound model (Extended Data Table 2). The align-pretrained PULSE model again significantly outperformed the MAE-pretrained RETFound model across all four disease end points, confirming that aligning retinal imaging with systemic physiological data yields a more powerful and generalizable representation for cross-body disease risk prediction (Fig. 5i–l).

Discussion

PULSE differs from existing longitudinal clinical modeling approaches in its core objective. Most prior methods apply general sequence architectures (RNNs and transformers) to clinical trajectories, typically focusing on a single modality. Multimodal extensions usually rely on input-level concatenation or late fusion of independently encoded representations, with representation learning driven by diagnostic labels. These approaches inadequately model how modalities biologically couple and covary as physiological states evolve. By contrast, PULSE explicitly aligns information across visits and modalities in a shared latent space, allowing modalities to provide mutual supervisory constraints. By jointly leveraging longitudinal context and cross-modal covariation, PULSE improves robustness under realistic missingness and supports reliable inference in heterogeneous clinical data.

A further future direction is the development of patient-specific longitudinal models. Although PULSE currently learns shared temporal and cross-modal structure at the population level, its design also suggests a natural path toward more individualized modeling. In particular, once longitudinal data become sufficiently dense within each patient, for example, with more follow-up visits, richer multimodal coverage and longer observation windows, it may become possible to train or fine-tune patient-specific models that more precisely capture unique physiological trajectories. Such extensions could allow the model to distinguish stable personal baselines from individual-specific rates and directions of change. Future work may therefore explore patient-level adaptation strategies built on PULSE, including patient-specific fine-tuning or hierarchical formulations that combine shared population structure with personalized trajectory modeling.

The PULSE framework carries potential to alleviate healthcare burdens and advance health equity[32]. By generating actionable molecular profiles from inexpensive, routine data, it democratizes access to advanced phenotyping, which is typically confined to large-scale research biobanks. This capability is particularly impactful for resource-limited settings, where dense multiomics profiling is prohibitively costly. Remarkably, the generated proteomic profiles achieve predictive accuracy comparable to ground-truth measurements. In some instances, models using generated data even outperformed those using original inputs, suggesting that the generative process acts as a powerful representational optimizer, distilling and refining biological signals from sparse clinical data[33]. Consequently, this cost-effective strategy can help mitigate health disparities and integrate precision medicine tools into routine clinical care[34].

PULSE is intended as a general framework rather than a universal model. Its value lies in providing a principled, flexible approach to develop task-specific models across modality combinations and clinical settings. Two limitations should be acknowledged. First, the specific models developed here should not be assumed to generalize across all populations; robustness across demographic subgroups and healthcare systems remains to be established. Second, PULSE should be viewed as a proof-of-concept framework requiring rigorous task-specific evaluation and prospective clinical validation before broad deployment.

Methods

The work was conducted in compliance with the Chinese Center for Disease Control and Prevention (CDC) policy on reportable infectious diseases and the Chinese Health and Quarantine Law and in compliance with patient privacy regulations in China and was adherent to the tenets of the Declaration of Helsinki. Ethics Committee approvals were obtained in all following mentioned institutions from the China Health Aging Investigation (CHAI), as a project of the International Consortium of Digital Twin in Medicine[35].

Datasets

For the UK Biobank Cohort[36], the UK Biobank recruited approximately 500,000 participants aged 40–69 years from across the UK between 2006 and 2010. This resource provides deeply phenotyped data, and for our study, we accessed several integrated datasets for each participant. These included multiomics data from blood samples (laboratory tests, proteomics and metabolomics), medical history, retinal fundus images and longitudinal diagnostic records for disease outcomes.

For the Systemic Diseases Prediction Cohort (a part of the CHAI project), we collected a longitudinal adult cohort from The First Affiliated Hospital of Wenzhou Medical University and the Eye Hospital of Wenzhou Medical University in Wenzhou, China. The cohort consisted of 127,731 adult participants who contributed a total of 657,103 clinical visits. For each participant, baseline demographic information was available. Each longitudinal visit record contained either retinal fundus photography, a panel of laboratory test results or both. These records were linked to follow-up data, including International Classification of Diseases (ICD)-based diagnostic codes, to track systemic diseases over time.

For the Myopia Prediction Cohort (a part of the CHAI project), we collected a longitudinal children cohort from The First Affiliated Hospital of Wenzhou Medical University and the Eye Hospital of Wenzhou Medical University in Wenzhou, China. This cohort consisted of 24,605 children who contributed a total of 50,043 visits. For each participant, baseline demographic information was available. Each longitudinal visit record contained either retinal fundus photography, a panel of laboratory test results or both. These records were linked to follow-up data, including AL and SE to track the development and progression of myopia.

For the Wenzhou ICU Cohort (a part of the CHAI project), we collected a longitudinal adult cohort from The First Affiliated Hospital of Wenzhou Medical University in Wenzhou, China. From the adult population (≥18 years old), we screened and enrolled 13,816 patients who were admitted to the ICU and survived the hospitalization. These patients contributed a total of 13,816 ICU admissions (first admissions), of which 739 cases resulted in 30-day readmission. Each ICU stay

included structured laboratory tests and unstructured clinical notes recorded during the first 48 h. These records were linked to follow-up data to track 30-day postdischarge readmission outcomes.

PULSE architecture and components

PULSE is a variational generative framework for longitudinal multimodal clinical data, extended with a modular temporal state-propagation component. PULSE models each patient as an evolving latent physiological process and can flexibly implement temporal propagation using either an RNN-like update or a transformer-like causal attention module, depending on the application setting.

(1) Modality-specific encoders and Past-State integration: for patient i at visit t, let $x_{i,t}^{(m)}$ denote modality m measured at the current visit, and let $h_{i,t}$ denote the patient-specific historical state summarized from all previous visits before t. For each currently available modality $m \in \mathcal{O}_{i,t}$, a modality-specific variational encoder maps the input to a diagonal Gaussian posterior in a shared latent space

$$q_{\phi_m}\left(z_{i,t}^{(m)}, |, x_{i,t}^{(m)}\right) = \mathcal{N}\left(\mu_{i,t}^{(m)}, \mathrm{diag}\left(\left(\sigma_{i,t}^{(m)}\right)^2\right)\right).$$

The historical state is processed by a dedicated Past-State encoder

$$q_{\phi_P}\left(z_{i,t}^{P}, |, h_{i,t}\right) = \mathcal{N}\left(\mu_{i,t}^{P}, \mathrm{diag}\left(\left(\sigma_{i,t}^{P}\right)^2\right)\right)$$

PULSE is motivated by the assumption that different clinical modalities are partial and noisy manifestations of a shared latent physiological state that evolves over time within each individual. Under this view, the historical state provides individualized physiological context that constrains biologically plausible cross-visit transitions. Consequently, conditioning on $h_{i,t}$ makes cross-modal completion at the current visit better posed than reconstruction from a single contemporaneous modality alone.

(2) Joint latent space, soft alignment and Visit-State estimation: the goal of PULSE is to integrate the current-visit modality representations and the historical context into a unified latent Visit-State. To encourage the latent branches to capture shared physiological structure while preserving current-visit variation, PULSE uses a distributional alignment strategy. For any two diagonal Gaussian posteriors $q_a = \mathcal{N}\left(\mu_a, \mathrm{diag}\left(\sigma_a^2\right)\right)$ and $q_b = \mathcal{N}\left(\mu_b, \mathrm{diag}\left(\sigma_b^2\right)\right)$, the Kullback–Leibler (KL) divergence is defined as

$$D_{\mathrm{KL}}\left(q_a | q_b\right) = \frac{1}{2}\sum_{d=1}^{D}\left[\log\left(\frac{\sigma_{b,d}^2}{\sigma_{a,d}^2}\right) + \frac{\sigma_{a,d}^2 + \left(\mu_{a,d} - \mu_{b,d}\right)^2}{\sigma_{b,d}^2} - 1\right].$$

The Jeffrey divergence is then given by

$$J\left(q_a, q_b\right) = D_{\mathrm{KL}}\left(q_a | q_b\right) + D_{\mathrm{KL}}\left(q_b | q_a\right)$$

Let $\mathcal{B}_{i,t} = \{q_{\phi_m} : m \in \mathcal{O}_{i,t}\} \cup \{q_{\phi_P}\}$ denote the set of latent branches available at visit t, including the Past-State branch. The alignment penalty is defined as the average pairwise Jeffrey divergence across all unordered branch pairs

$$\mathcal{L}_{align}^{(i,t)} = \frac{1}{|\mathcal{P}_{i,t}|}\sum_{(a,b)\in\mathcal{P}_{i,t}} J\left(q_a, q_b\right)$$

where $\mathcal{P}_{i,t}$ is the set of all unordered pairs drawn from $\mathcal{B}_{i,t}$. This distributional alignment encourages biological semantic consistency across branches without forcing exact geometric identity. In this way, the historical state acts as a supportive individualized context rather than as a rigid template, allowing the model to preserve meaningful current-visit-specific variation.

The integrated Visit-State is then computed by equal-weight fusion of the posterior means of the currently available modalities and the Past-State branch,

$$z_{i,t} = \frac{\mu_{i,t}^{P} + \sum_{m\in O_{i,t}} \mu_{i,t}^{(m)}}{|O_{i,t}| + 1}$$

Importantly, the denominator counts only the currently available modalities and does not include the Past-State branch; therefore, the historical state is incorporated exactly once and is not double counted. If no current modality is available, the Visit-State defaults to the Past-State representation, that is, $z_{i,t} = \mu_{i,t}^{P}$.

(3) Temporal propagation and modality-specific decoders: after estimating the Visit-State $z_{i,t}$, PULSE propagates it forward to construct the historical context for the next visit. In the recurrent instantiation, the temporal module updates the historical state by a gated recurrent transition

$$h_{i,t+1} = f_{\mathrm{temp}}\left(h_{i,t}, z_{i,t}\right).$$

In the transformer-like instantiation, the historical state is computed from the causally masked sequence of previous Visit-States $(z_{i,1}, \ldots, z_{i,t})$. In both cases, the temporal module serves the same role: summarizing prior patient-specific latent states into the Past-State consumed at the next visit. The Visit-State is then passed to modality-specific decoders to reconstruct all target modalities

$$\hat{x}_{i,t}^{(m)} = g_{\theta_m}\left(z_{i,t}\right), \forall m.$$

This design enables PULSE to place visits with complete and incomplete measurements in a common latent space and to impute missing modalities from the fused latent physiological representation.

PULSE uses modality-specific variational encoders that project heterogeneous inputs into a shared 16-dimensional latent space. For tabular modalities, proteomics (2,923 features), metabolomics (251 features), laboratory tests (61 features) and patient context, a two-layer nonlinear neural network directly serves as the variational encoder, mapping the input features to a 16-dimensional isotropic multivariate normal distribution. For medical history, all historical medical concept codes are first segmented into nonoverlapping 5-year windows and embedded using a word2vec model into 400-dimensional vectors. Within each window, these embeddings are averaged to obtain a single representation, and the resulting time-ordered sequence of window embeddings is processed by a RNN to produce a history summary for the current visit; this summary is then passed through a variational encoder that outputs a 16-dimensional isotropic multivariate normal distribution. We also provided a text-based representation of longitudinal disease history. In this setting, historical diagnosis records were converted into chronological natural-language descriptions and encoded using a general large language model (for instance, Llama 3) to obtain embeddings as an alternative patient-context representation.

The training and inference procedure was as follows:

(1) Variational objective: PULSE is trained by minimizing a variational objective composed of reconstruction, KL regularization, alignment and adversarial terms

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_{\mathrm{KL}}\alpha\mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{align}}\mathcal{L}_{align} + \lambda_{\mathrm{adv}}\mathcal{L}_{adv},$$

where λ_{KL}, λ_{align} and λ_{adv} are scalar hyperparameters, and $\alpha \in [0,1]$ is a KL annealing weight that is gradually increased during early training to improve optimization stability and reduce posterior collapse. The KL term regularizes both the modality-specific latent branches and the Past-State branch toward a standard normal prior $p(z) = \mathcal{N}(0, I)$.

(2) Reconstruction loss and active masking: to prevent PULSE from relying on trivial within-visit shortcuts, we use an active masking strategy during training. At a visit with more than one observed modality, one currently observed modality is randomly selected and actively masked with probability p_{mask}. The actively masked modality is removed from the encoder input, but its ground-truth measurement is still used as a reconstruction target. This forces the model to leverage the remaining current modalities together with the historical state for reconstruction. Let $s_{i,t}^{(m)} \in \{0,1,2\}$ denote the status of modality m at visit t, where $s = 0$ indicates observed and used as input, $s = 1$ indicates actively masked during training and $s = 2$ indicates naturally missing. The reconstruction loss is defined as

$$\mathcal{L}_{rec} = \sum_{i,t} \sum_{m} 1\left[s_{i,t}^{(m)} \in \{0,1\}\right] l\left(\hat{x}_{i,t}^{(m)}, x_{i,t}^{(m)}\right),$$

where $l(\cdot,\cdot)$ is the MSE.

Thus, modalities that are naturally missing do not contribute supervision, whereas observed and actively masked modalities both contribute to the reconstruction objective whenever ground-truth measurements are available. This design anchors the latent representation to the current visit while discouraging the model from ignoring the historical context.

(3) Missingness adversarial training: to reduce information leakage from modality missingness patterns, we introduce a modality-wise adversarial regularizer. For each modality m, an adversary D_m receives the reconstructed modality $\hat{x}_{i,t}^{(m)}$ and predicts whether that modality was missing at the current visit. The adversarial loss is

$$\mathcal{L}_{adv} = \sum_{i,t} \sum_{m} \text{BCE}\left(D_m\left(\hat{x}_{i,t}^{(m)}\right), y_{i,t}^{(m)}\right)$$

where BCE denotes binary cross-entropy. A gradient reversal layer is inserted before each adversary so that, while the adversary is trained to detect missingness, the reconstruction pathway is simultaneously encouraged to remove missingness-specific cues from the generated outputs. This regularization helps PULSE learn reconstructions that reflect physiological structure rather than artifacts induced by the missing-data pattern.

The model parameters were optimized using the AdamW optimizer with a learning rate of 1×10^{-4}, a weight decay of 1×10^{-3} and a mini-batch size of 64 to ensure stable convergence. We adopted a threefold cross-validation strategy for robust performance assessment, training the model for a maximum of 500 epochs. To prevent overfitting and reduce computational cost, an early stopping mechanism was used to terminate training if the reconstruction loss on the validation dataset showed no improvement over a patience period of ten epochs, with the model checkpoint yielding the optimal validation performance retained for subsequent evaluation. Furthermore, to facilitate the effective learning of the latent space and mitigate posterior collapse, we implemented a KL divergence annealing schedule, wherein the KL term penalizing the deviation of the latent representation from the prior was down-weighted during the initial 50 epochs of the training process.

Adaptation of single-cell multimodal frameworks

The clinical dataset consists of measurements from 10,000 UK Biobank participants, involving 61 routine laboratory tests (modality A) and 251 metabolomics biomarkers (modality B). To prepare the data for single-cell multimodal frameworks, which are typically designed for count-based genomic data, we first performed log-transformation followed by Z-score normalization on all continuous clinical variables to ensure they are on a comparable scale. The dataset was split at the patient level into a training set (70%, $n = 7{,}000$) and a test set (30%, $n = 3{,}000$). For the single-cell frameworks, each visit (Baseline and Follow-up) from the training participants was treated as an independent paired sample (A and B), effectively converting the longitudinal data into a static dataset of 14,000 paired observations for training.

We adapted three representative single-cell frameworks to handle continuous clinical data and perform cross-modality imputation:

(1) MIDAS adaptation: MIDAS is a deep probabilistic framework based on a product-of-experts (PoE) architecture. We adapted the input and output layers of MIDAS to handle continuous feature blocks instead of discrete counts. The negative binomial likelihood typically used for single-cell data was replaced with a Gaussian likelihood to model clinical biomarkers. The framework uses modality-specific encoders to map laboratory tests and metabolomics into a shared latent space. The PoE mechanism allows the model to generate a joint posterior distribution even when one modality is missing during inference.

(2) scVAEIT adaptation: scVAEIT uses a variational autoencoder with a missing mask mechanism. We adapted this by concatenating the laboratory tests and metabolomics into a single vector. We configured the model to use a continuous Gaussian distribution for the reconstruction loss. The core adaptation involves the masking strategy: during training, the model learns the conditional distribution of metabolomics given laboratory tests by randomly masking features and minimizing the reconstruction error of the unobserved variables.

(3) StabMap adaptation: StabMap is a nongenerative approach that uses mosaic data topology and mutual-nearest-neighbor (MNN) alignment. We designated the paired training data as the reference dataset. The 61 laboratory tests served as the shared features between the training reference and the test query. StabMap builds a topology where the test samples are projected onto the reference coordinates by traversing the shortest path defined by the shared laboratory features, enabling the imputation of the unshared metabolomics features on the basis of the reference distribution.

For the training procedure, all three adapted frameworks were trained using the 7,000 training-set individuals. Each visit was treated as an independent sample, and the models were trained without access to any temporal or historical state variables. For MIDAS and scVAEIT, training involved optimizing the evidence lower bound (ELBO), which includes the reconstruction accuracy of both the laboratory tests and the metabolomics biomarkers alongside a regularization term for the latent space. StabMap was trained by establishing the reference coordinate system and embedding the training data into a stabilized mosaic space. No information from the test set was seen during this phase.

For testing and evaluation, at the test time, the evaluation was standardized to match the target task of reconstructing Follow-up Visit metabolomics from Follow-up Visit laboratory tests. For each held-out patient in the test set ($n = 3{,}000$), only the laboratory tests from their second visit were provided as input to the models. MIDAS and scVAEIT performed inference by passing the laboratory tests through their trained encoders and using the resulting latent representations to decode/impute the missing metabolomics. StabMap projected the test laboratory tests onto the reference space and retrieved the corresponding metabolomics values. All models were evaluated by comparing the reconstructed metabolomics against the ground-truth Follow-up Visit measurements using metrics such as MSE and Pearson correlation. Unlike the longitudinal PULSE framework, these static frameworks relied solely on the current visit's laboratory data for reconstruction.

Benchmarking against existing longitudinal clinical modeling methods

To position PULSE relative to representative longitudinal clinical modeling paradigms, we additionally benchmarked it on the plasma proteomics generation task using longitudinal clinical history and current routine laboratory tests as input. This benchmark was conducted on the UK Biobank-PPP cohort, in which each sample consisted of all medical history recorded before the current visit, the current-visit routine laboratory measurements and the corresponding measured plasma proteomics profile. Data were split at the patient level into development and held-out test sets, such that no individual contributed records to both training and testing. The development set was further used for model training and validation. For all compared methods, the prediction target was identical: generation of the current-visit proteomics vector.

We compared PULSE with three representative paradigms from longitudinal clinical data modeling: a Delphi-style diagnosis-trajectory model, an adapted Llama 3.1 text-embedding model and a standard Transformer encoder. These methods were selected to represent three common strategies for modeling longitudinal patient information: sequence modeling of diagnostic event trajectories, natural-language-based patient summarization using a general-purpose large language model and direct token-based encoding of longitudinal structured clinical data.

For the Llama 3.1 baseline, all precurrent-visit diagnosis records were converted into chronological natural-language descriptions and current-visit routine laboratory measurements were appended to the same patient summary in text form. Laboratory values were described using both their measured continuous values and discretized levels. These patient summaries were encoded using a frozen Llama 3.1 model, and the resulting text embeddings were passed to a multilayer regression head to predict the current-visit proteomics vector. This baseline tests whether general-purpose text embeddings of longitudinal clinical context are sufficient for proteomics generation without explicit cross-modal alignment.

For the standard Transformer baseline, longitudinal diagnosis history and current routine laboratory data were represented as a structured token sequence. Diagnosis codes were encoded as age-aware event tokens, and current laboratory measurements were discretized into feature-specific bins and encoded as laboratory tokens. Token embeddings were combined with type embeddings, age embeddings and positional embeddings and then processed by a Transformer encoder. The final (CLS) representation was passed to a regression head to predict the proteomics vector. CLS is the classification token in transformer, the aggregated sequence representation used for the prediction head. This model directly uses all available structured inputs but does not impose an explicit alignment mechanism across modalities.

For the Delphi-style baseline, we adapted a diagnosis-trajectory modeling framework to the same prediction task by using longitudinal disease-event history before the current visit to derive a patient trajectory representation, which was then passed to a regression head for proteomics prediction. As this class of model is designed primarily for longitudinal disease-event trajectories rather than multimodal completion, it could not natively incorporate current routine laboratory measurements in the same way as PULSE. It therefore served as a diagnosis-history-only longitudinal trajectory baseline in this benchmark.

To ensure a fair comparison, all methods were trained and evaluated under the same patient-level split and against the same current-visit proteomics target. The differences between methods therefore reflect their underlying representational assumptions rather than differences in the prediction end point. In particular, PULSE explicitly models current laboratory data and historical context as complementary branches within a shared latent physiological space, whereas the comparator methods rely on event-sequence modeling, text embedding or direct token-sequence encoding without explicit cross-modal alignment.

Benchmark scripts and implementation details for these longitudinal clinical modeling comparisons are provided in the public benchmark repository.

Biological aging trajectories and grouping strategy

Trajectory inference was performed directly on the PULSE latent representations. A *k*-nearest neighbor graph was constructed, followed by diffusion pseudotime computation. Using CellRank[30,31], a pseudotime-informed transition matrix (PseudotimeKernel, time_key="dpt_pseudotime") was generated to derive a vector field of transition probabilities in latent representations. Streamlines representing long-range directional flows were obtained via numerical integration of this vector field and subsequently projected onto a UMAP space (n_neighbors = 30, min_dist = 0.5 and random_state = 42) for visualization.

Trajectory groups were defined to isolate trajectory effects from chronological age. Participants were binned into 0.2-year age intervals; within each bin, individuals were ranked by UMAP2 coordinate and partitioned into three equal-sized tertiles (upper, middle and lower). Assignments were pooled across all age bins to form three trajectory groups spanning the full age range. Clinical validation used sex-stratified Kaplan–Meier survival analysis (lifelines), with age (years) as the time variable and trajectory group as the stratification factor, assessing cumulative incidence of major age-related diseases.

Multicenter 30-day readmission among ICU patients

To evaluate the generalizability and clinical robustness of the PULSE framework across diverse healthcare systems, we constructed a longitudinal multimodal benchmark using two independent cohorts: the MIMIC-III database (US-based; Supplementary Table 1) and a large-scale ICU dataset from Wenzhou, China (Wenzhou ICU cohort; Supplementary Table 1). The Wenzhou ICU cohort is a part of The CHAI, as a project of the International Consortium of Digital Twin in Medicine. Data for this study were sourced from several hospitals in the CHAI project. Ethics Committee approvals were obtained for CHAI. The study was registered at clinicaltrial.gov (NCT06791486). The work was conducted in compliance with the Chinese CDC policy on reportable infectious diseases and the Chinese Health and Quarantine Law, in compliance with patient privacy regulations in China, and was adherent to the tenets of the Declaration of Helsinki. We focused on the prediction of 30-day hospital readmission, a task that requires modeling complex physiological trajectories and postdischarge risk.

Data extraction and standardized processing. For both cohorts, we implemented a standardized data processing pipeline to ensure consistency. The prediction start ($T = 0$) was defined as the timestamp of the patient's first ICU entry (intime). We used an observation window of the first 48 h following entry, which was divided into 12 nonoverlapping 4-h bins:

(1) Structured modality: within each 4-h bin, measurements were summarized from vital signs and laboratory variables using six statistics (mean, median, maximum, minimum, count and variance) to create a fixed-length longitudinal representation.
(2) Text modality: time-stamped clinical notes recorded within each 4-h bin were concatenated and encoded at the step level using a frozen general large language model (Llama-3.1-8B) to obtain embeddings.
(3) Label definition: the 30-day readmission task was defined on the basis of the hospital discharge time (dischtime). Admissions were labeled positive if a subsequent hospital admission occurred within 30 days of discharge. Following standard protocols, we excluded admissions involving in-hospital death, newborn types or stays shorter than 24 h to focus on adult patients with sufficient longitudinal data.

Samples were split at the patient level into training, validation and test sets (7:1:2 ratio) to ensure the model generalizes to entirely unseen individuals. Binary classifiers were optimized using class-weighted logistic loss to account for the low incidence rate of readmission.

Baseline comparisons. We compared PULSE against three established longitudinal modeling paradigms:

- Transformer (laboratory tests): a structured-only baseline using a Transformer encoder over the 12-step sequence of summary statistics
- LLM (clinical notes): a text-only baseline where all notes from the observation window were merged, encoded via Llama-3.1-8B and passed to a multilayer perceptron
- Directly Combined (late fusion): this model encoded structured and text modalities separately using the encoders above and combined their representations only at the final prediction stage

PULSE implementation. Unlike the late-fusion baseline, PULSE preserved multimodal alignment at the 4-h-step level. At each time step, the framework aligned the latent distributions of structured measurements and text embeddings while simultaneously conditioning on a Past-State propagated from previous steps. This design allowed PULSE to capture within-step multimodal correspondence and across-step physiological progression in a unified latent space.

Plasma proteomic disease prediction modeling

For each disease, a prediction model was developed using a LightGBM. The objective of the model was to classify whether a participant, healthy at baseline, would develop the specified disease (class 1) or remain healthy (class 0) within a 10-year follow-up period; all individuals with a prevalent diagnosis were excluded. The LightGBM models were trained on the training set using the true plasma proteomic features combined with age and sex as input. Testing was performed on the held-out test set by comparing two input scenarios: using the generated plasma proteomic features combined with age and sex and using the true plasma proteomic features combined with age and sex. A comparison was also conducted by developing a separate LightGBM model trained on the combined inputs of routine laboratory tests, the encoded medical history context vector as previous, age and sex. This model directly assesses the predictive power, without translating into proteomic profile.

Baseline methods of multimodal alignment

For cross-modal alignment pretraining for fundus images and laboratory tests, PULSE was pretrained on longitudinal mosaic data from the development set, where each visit could contain fundus imaging, laboratory tests, or both. Fundus images were first processed through a RETFound-pretrained encoder to extract high-level features. These fundus-derived features were then aligned with laboratory test measurements and the patient's past state (derived from previous visits) in a joint latent space using the PULSE modality-specific encoder and alignment framework.

For systemic disease prediction fine-tuning, after alignment pretraining, latent state representation of visits with fundus images was fine-tuned for disease prediction using only fundus images as input. For each disease (ischemic stroke, myocardial infarction, heart failure and dementia), we constructed a binary classification task for 5-year incident disease prediction. Visits with pre-existing disease were excluded. A visit was labeled as positive if the disease occurred within 5 years of follow-up and negative otherwise. In single-visit mode, the fundus encoder produced a latent representation from the current visit only, which was fed to a classification head. In multivisit mode, fundus images from the current and all previous visits were sequentially processed, with the final visit representation aggregating the longitudinal trajectory before prediction. The model was trained using binary cross-entropy loss on the development set and evaluated on the held-out test set under both prediction regimes.

For the baseline of systemic disease prediction comparison, the RETFound baseline used MAE-pretrained RETFound encoder features directly. For single-visit prediction, fundus features from the current visit were passed to a classification head. For multivisit prediction, an RNN aggregated features from the current and historical fundus images before classification. Both PULSE and RETFound models were trained and evaluated on identical disease tasks and prediction regimes for fair comparison.

For myopia progression prediction fine-tuning, after alignment pretraining, the PULSE laboratory test encoder was fine-tuned for myopia progression prediction using only laboratory tests as input. The task was to predict future AL and SE trajectories over 1–5-year horizons. For each visit with available laboratory tests, the encoder processed the current visit's laboratory tests combined with the patient's past state (aggregated from previous visits), producing a latent representation that was fed to a regression head to predict future refractive measurements at 1, 2, 3, 4 and 5 years. The model was trained using MSE loss on the development set and evaluated on the held-out test set. Performance was assessed using the coefficient of determination (R^2), MAE and Pearson correlation coefficient between predicted and actual values at each future time point.

For baseline comparison, the end-to-end baseline was trained solely on laboratory tests without cross-modal pretraining. Laboratory tests from the current visit and historical visits were processed through a recurrent architecture to aggregate longitudinal information, and the resulting representation was passed to a regression head to predict future AL and SE trajectories, using the same loss function and evaluation metrics as the PULSE model for fair comparison.

Statistics and reproducibility

Study design and sample size. This study employed a retrospective observational design using existing longitudinal cohort data. No statistical method was used to predetermine sample size. Sample sizes were determined by data availability in each cohort. For the UK Biobank metabolomics-imputation task, we included ~10,000 participants with paired baseline and follow-up measurements. The UK Biobank-PPP proteomics cohort comprised 50,000 participants. The MIMIC-III ICU cohort included 44,265 adult patients, and the Wenzhou ICU cohort included 13,816 adult patients. The myopia prediction cohort included 24,605 children contributing 50,043 visits, and the systemic disease prediction cohort included 127,731 adults contributing 657,103 visits.

Data exclusion. In the ICU readmission analysis, we excluded admissions involving in-hospital death, newborn types or ICU stays shorter than 24 h, following standard protocols to focus on adult patients with sufficient longitudinal data. For disease prediction tasks, individuals with a prevalent diagnosis at baseline were excluded to enable incident disease prediction within a 10-year follow-up window. For myopia progression prediction, visits with pre-existing high myopia were excluded to track the development and progression of refractive changes. No other data were excluded from the analyses.

Randomization and data splitting. The experiments were not randomized. For model development and evaluation, data were partitioned at the patient level (not the visit level) to ensure that all records from any given individual appeared in only one split. In the UK Biobank metabolomics task, individuals were randomly assigned to a 70% training set (n = 7,000) and a 30% held-out test set (n = 3,000). The systemic disease and myopia cohorts were randomly split at the individual level into 70% development and 30% held-out test sets. The ICU cohorts were randomly partitioned at the patient level into training, validation and

test sets in a 7:1:2 ratio. Random seeds were fixed where specified (for example, UMAP visualization used a random state of 42).

Blinding. The investigators were not blinded to allocation during experiments and outcome assessment. This is a computational modeling study using retrospective observational data; all data splits and evaluation procedures were prespecified in the analysis protocol.

Statistical analyses. Correlation analyses were performed using Pearson's correlation coefficient (r) for linear relationships and Spearman's rank correlation coefficient (ρ) for monotonic relationships, as specified in the figure legends. The statistical significance of correlations and group differences was assessed using two-sided t-tests or Wilcoxon signed-rank tests, depending on the normality of the data distribution. For benchmarking comparisons involving paired predictions (for example, PULSE versus baseline methods), the two-sided Wilcoxon signed-rank test was applied across biomarkers or proteins.

95% CIs for model performance metrics (AUROC and AUPRC) and regression coefficients were estimated using nonparametric bootstrap with 200 resamples, unless otherwise stated. In the main text and figures, exact P values are reported where $P \geq 0.0001$; values below this threshold are denoted as $P < 0.0001$. For most comparative analyses, no adjustment for multiple comparisons was applied; exceptions or specific adjustments are noted in the relevant figure legends.

Trajectory inference was performed using CellRank (v2)[31]. We embedded unpaired multimodal data from the UK Biobank into a unified latent space and computationally inferred aging progression trajectories using pseudotemporal ordering. Cumulative incidence curves were calculated using the Kaplan–Meier estimator via the lifelinesmodule (Python). Pairwise group differences in survival distributions were assessed using two-sided log-rank tests.

For disease prediction tasks using LightGBM, models were trained on the development set and evaluated on the held-out test set. Class imbalance was addressed using class-weighted logistic loss during training. Hyperparameters were tuned via threefold cross-validation within the training set.

Reproducibility. All source code for model implementation, training and evaluation is publicly available at https://github.com/ww20hust/LongitudinalGeneration. The UK Biobank data are available through the UK Biobank access procedure[37]. The MIMIC-III data are available via PhysioNet (https://physionet.org/). Restrictions apply to the myopia prediction cohort and Wenzhou ICU cohort (CHAI). Data access requests should be addressed to the corresponding authors and forwarded to a data access committee for approval. All experiments were conducted using fixed software environments and package versions as documented in the repository. Model checkpoints yielding optimal validation performance were retained for subsequent evaluation.

Computational cost

For the UK Biobank metabolomics-generation task, training can be completed in less than 1 h on a single modern graphics processing unit (GPU) with 24 GB memory. For the UK Biobank proteomics-imputation task, training can be completed in less than 3 h on the same hardware. In both settings, inference does not require a GPU and takes less than 1 s per patient on a standard laptop.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The MIMIC-III analyzed in this study is available via PhysioNet at https://physionet.org/. UKB data are available through a procedure described at https://www.ukbiobank.ac.uk/enable-your-research. The full UKB protocol is available at https://www.ukbiobank.ac.uk/media/gnkeyh2q/study-rationale.pdf. Restrictions apply to the availability of data from the CHAI project (Systemic Diseases Prediction Cohort, myopia prediction cohort and ICU cohort), which were used with the permission of the participants for the current study. Data access requests should be addressed to the corresponding authors and forwarded to a data access committee for approval.Source data are provided with this paper.

Code availability

The code for using PULSE, as well as the benchmark, is available via GitHub at https://github.com/ww20hust/LongitudinalGeneration and via Zenodo at https://doi.org/10.5281/zenodo.20555139 (ref. 38).

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Author contributions

W.W., G.L., K.W., H. Xu, H. Xiao, C.H., S.L., C.T., F.L., Z.Z., B.L., J.L., C.L.Z., H.W., I.C., W.L., Z.S., Y.Y., A.L., E.O., S.A.A.D., H.Z., J.G., O.O.A., V.N.G., J.E.J.R. and K.Z. collected and analyzed the data. K.Z. conceived, designed and supervised the project. K.Z., J.E.J.R., G.L., W.W. and K.W. wrote the paper. All authors discussed the results and reviewed the manuscript.

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Competing interests

W.W., K.W. and K.Z. have filed patents related to this work (jurisdiction: international, PCT – Patent Cooperation Treaty; application no. PCT/CN2025/112123, filed with the China National Intellectual Property Administration as receiving office). All other authors declare no competing interest.

Additional information

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Extended Data Table 1 | Demographic table of the cohort for systemic diseases risk prediction

Disease	Patients (M/F)	Visits (total)	Visits (Fundus)	Visits (lab test)	Visits (dual)
Dementia	810(419/391)	9775	962	8416	397
Heart failure	2233(1199/1034)	29578	2749	25768	1061
MI	5216(2765/2451)	65123	7145	55597	2381
IS	7905(4100/3805)	89936	10985	75586	3365
Control	111567(50081/61486)	462691	144916	3365	64707

Each row corresponds to a disease (ischemic stroke [IS], myocardial infarction [MI], heart failure, dementia, or control), listing the number of patients (male/female [M/F]), total visits, and visits categorized by data type: fundus-only (retinal fundus photography only), lab test-only (routine laboratory test data only), and dual (both fundus and laboratory test data).

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Extended Data Table 2 | Demographic table of the UK biobank cohort

Disease	Total Cases	Biochemistry	Proteomics	Metabolomics	Retinal Image
Anemia	41735	39084	3774	22541	
Diabetes	25653	23744	2236	13583	
Dementia	11135	10421	1233	5931	1376
Parkinson disease	3803	3566	348	2040	
Hypertension	68689	63788	5840	36311	
MI	39772	37355	3495	21263	6021
Arrhythmias	41729	39407	3661	22392	
Heart failure	18608	17451	1801	10093	2595
Stroke	12997	12153	1154	6837	1799
PRD	19061	17875	1752	10354	
COPD	24255	22613	2188	13082	
Asthma	15500	14032	1294	7981	
DSD	126530	118003	10548	66569	
IBD	18013	16758	1592	9443	
Liver diseases	17252	16197	1506	9178	
Osteoarthritis	68907	64625	5822	36788	
Renal failure	41567	39146	3787	22350	
Total	502140	476569	42576	266380	

This table summarizes the total cases for 17 major diseases in the UK Biobank, along with the number of cases with available biochemistry, proteomics, metabolomics, and retinal image data. Abbreviations: MI, myocardial infarction; PRD, peripheral artery disease; COPD, chronic obstructive pulmonary disease; DSD, digestive system disorders; IBD, inflammatory bowel disease.

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☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
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☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No special software or code was used to collect the data.
Data analysis	Trajectory inference was performed using CellRank (v2). We used Pytorch for data analysis. Custom code was at https://github.com/ww20hust/LongitudinalGeneration

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project (Systemic Diseases Prediction Cohort, myopia prediction cohort and ICU cohort), which were used with the permission of the participants for the current study. Data access requests should be addressed to the corresponding authors and forwarded to a data access committee for approval.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	We used "sex" in our manuscript because we considered the biological attributes.
Reporting on race, ethnicity, or other socially relevant groupings	Information on race, ethnicity or other socially relevant groupings was not available for all included datasets.
Population characteristics	UK Biobank Cohort: The UK Biobank recruited approximately 500,000 participants aged 40-69 years from across the United Kingdom between 2006 and 2010. This resource provides deeply phenotyped data, and for our study, we accessed several integrated datasets for each participant. These included multi-omics data from blood samples (lab tests, proteomics, and metabolomics), medical history, retinal fundus images, and longitudinal diagnostic records for disease outcomes. Systemic Diseases Prediction Cohort (a part of CHAI project): We collected a longitudinal adult cohort from The First Affiliated Hospital of Wenzhou Medical University and the Eye Hospital of Wenzhou Medical University in Wenzhou, China. The cohort consisted of 127,731 adult participants who contributed a total of 657,103 clinical visits. For each participant, baseline demographic information was available. Each longitudinal visit record contained either retinal fundus photography, a panel of laboratory test results, or both. These records were linked to follow-up data, including ICD-based diagnostic codes, to track systemic diseases over time. Myopia Prediction Cohort (a part of CHAI project): We collected a longitudinal children cohort from The First Affiliated Hospital of Wenzhou Medical University and the Eye Hospital of Wenzhou Medical University in Wenzhou, China. This cohort consisted of 24,605 children who contributed a total of 50,043 visits. For each participant, baseline demographic information was available. Each longitudinal visit record contained either retinal fundus photography, a panel of laboratory test results, or both. These records were linked to follow-up data, including axial length (AL) and spherical equivalent (SE) to track the development and progression of myopia. Wenzhou ICU Cohort (a part of CHAI project): We collected a longitudinal adult cohort from The First Affiliated Hospital of Wenzhou Medical University in Wenzhou, China. From the adult population (≥18 years old), we screened and enrolled 13,816 patients who were admitted to the ICU and survived the hospitalization. These patients contributed a total of 13,816 ICU admissions (first admissions), of which 739 cases resulted in 30-day readmission. Each ICU stay included structured laboratory tests and unstructured clinical notes recorded during the first 48 hours. These records were linked to follow-up data to track 30-day post-discharge readmission outcomes. The work was conducted in compliance with the Chinese CDC policy on reportable infectious diseases and the Chinese Health and Quarantine Law and in compliance with patient privacy regulations in China and was adherent to the tenets of the Declaration of Helsinki. Ethics Committee approvals were obtained in all above mentioned institutions from the China Health Aging Investigation (CHAI), as a project of the International Consortium of Digital Twin in Medicine.
Recruitment	Participants have consented to use their EHR data for biomedical research. We used de-identified data in our study.
Ethics oversight	The study was performed in compliance with the tenets of the Declaration of Helsinki and was approved by Institutional Review Board/Ethics Committee of the participated hospitals.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Sample size	The size of the cohort was indicated in the tables.
Data exclusions	In the ICU readmission analysis, we excluded admissions involving in-hospital death, newborn types, or ICU stays shorter than 24 hours, following standard protocols to focus on adult patients with sufficient longitudinal data. For disease prediction tasks, individuals with a prevalent diagnosis at baseline were excluded to enable incident disease prediction within a 10-year follow-up window. For myopia progression prediction, visits with pre-existing high myopia were excluded to track the development and progression of refractive changes. No other data were excluded from the analyses.
Replication	All source code for model implementation, training, and evaluation is publicly available at https://github.com/ww20hust/

Replication	LongitudinalGeneration. The UK Biobank data are available through the UK Biobank access procedure (https://www.ukbiobank.ac.uk/enable-your-research). The MIMIC-III data are available via PhysioNet (https://physionet.org/). Restrictions apply to the myopia prediction cohort and Wenzhou ICU cohort (China Health Aging Investigation, CHAI). Data access requests should be addressed to the corresponding authors and forwarded to a Data Access Committee for approval. All experiments were conducted using fixed software environments and package versions as documented in the repository. Model checkpoints yielding optimal validation performance were retained for subsequent evaluation.
Randomization	Samples were randomly allocated to the training, tuning and testing sets.
Blinding	The Investigators were not blinded to allocation during experiments and outcome assessment. This is a computational modeling study using retrospective observational data; all data splits and evaluation procedures were pre-specified in the analysis protocol.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*